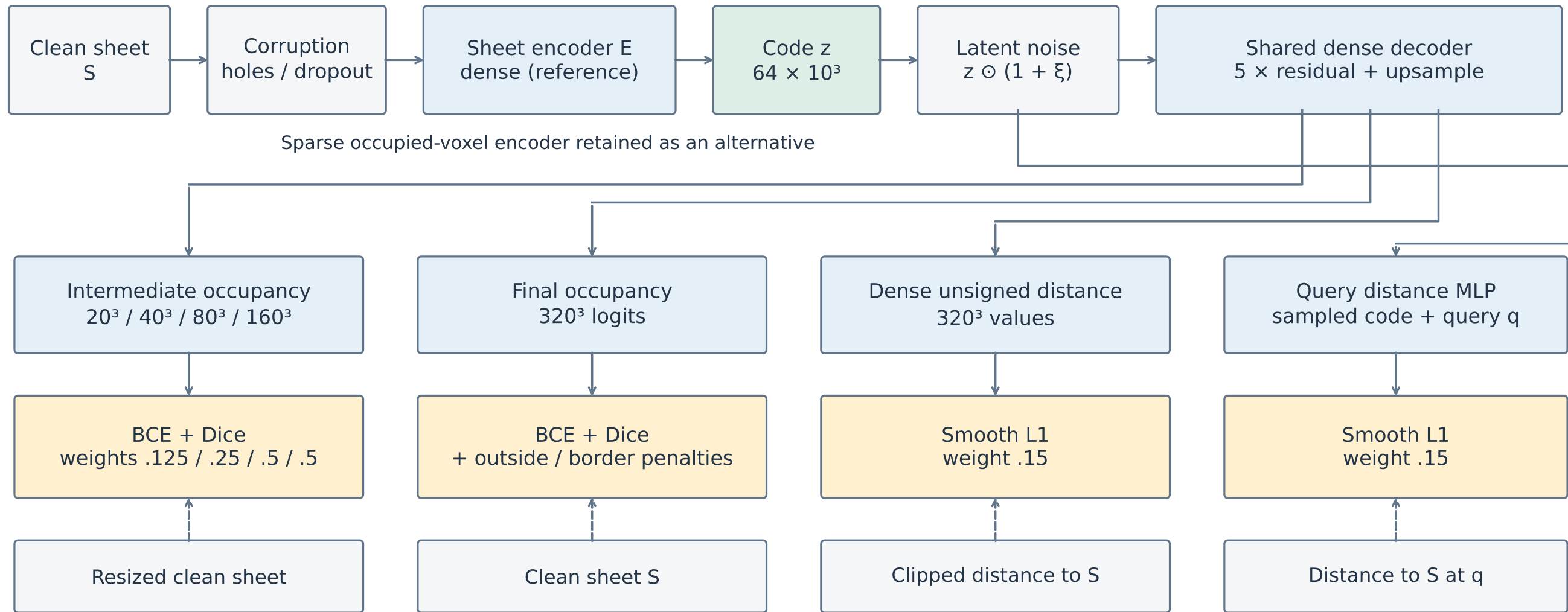


AE: learn a single-sheet code and freeze it for P2SD



Code regularization: distinct sheets in one crop → cosine repulsion on z (weight .1, margin .5).

All reconstruction heads depend on the bottleneck. No encoder-decoder skips; no explicit topology loss.